NETWORKING

VLANs
Visual Guide

Virtual LANs —
carving one switch into many networks.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/5

Broadcast Domain Segmentation

One physical switch, several logical networks that can't hear each other.

VLAN 10 — Eng

ports 1-8■broadcasts stay■here

VLAN 20 — Sales

ports 9-16■isolated from■VLAN 10

- A broadcast in VLAN 10 never reaches VLAN 20 — same isolation as separate switches, zero extra hardware.
- Security + sanity: HR, guests, IoT, and servers each get their own broadcast domain.
- VLANs are L2: devices in different VLANs need a ROUTER (L3) to talk — by design.

STEP 2/5

802.1Q Tagging

A 4-byte tag in the Ethernet header says which VLAN a frame belongs to.

```
[ dst MAC | src MAC | 802.1Q TAG | EtherType | payload | CRC ]

                              ■■ TPID 0x8100 + VLAN ID (12 bits: 1–4094)
```

- The tag is inserted between source MAC and EtherType — 12-bit VLAN ID = 4094 usable VLANs.
- Tags are added/removed at trunk ports; end hosts never see them (usually).
- Double-tagging (Q-in-Q) stacks tags for service providers carrying customer VLANs.

STEP 3/5

Access vs Trunk Ports

Edge ports speak untagged; uplinks carry everything tagged.

Access port

ONE vlan■untagged■end hosts:■PCs, printers

Trunk port

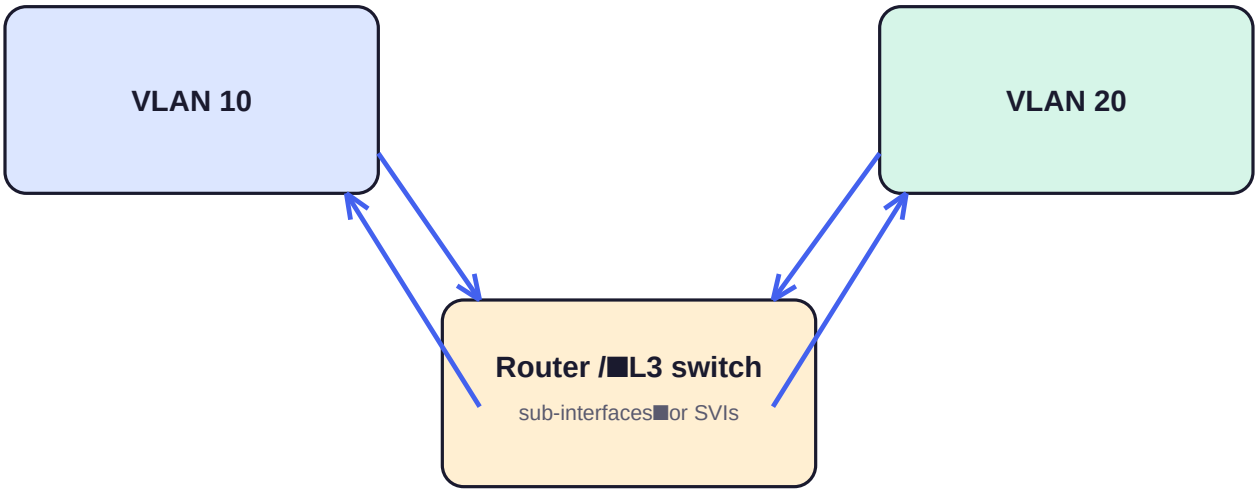
MANY vlans■tagged■switch↔switch■switch↔router

- Access: the switch assigns the VLAN — the device doesn't know or care.
- Trunk: frames carry tags so the far switch sorts them into the right VLANs.
- Native VLAN: the one untagged VLAN allowed on a trunk — keep it unused and prune it.

STEP 4/5

Inter-VLAN Routing

VLANs isolate — routers (or L3 switches) reconnect selectively.



Router-on-a-stick: one trunk to the router, sub-interfaces per VLAN. Modern: L3 switch with SVIs — routing at wire sp

STEP 5/5

VLAN Hopping & Hardening

Tags are trusted — misconfigurations are the attack.

- Double-tagging attack: crafted frames hop to another VLAN — defeated by proper native-VLAN hygiene.
- Never leave ports in trunk-auto: explicitly set access vs trunk; disable unused ports.
- Prune trunks: carry only the VLANs each link actually needs.
- Voice VLANs: phones tag voice traffic separately — QoS for calls without a second cable.

VLANs segment, they don't encrypt: traffic between VLANs is still plaintext on the wire without additional protection.

STEP ✓/✓

Key Takeaways

VLANs — Remember

- VLANs split one switch into isolated broadcast domains — security without extra hardware.
- 802.1Q inserts a 4-byte tag (VLAN ID 1–4094) into the Ethernet header.
- Access ports = one untagged VLAN (hosts); trunk ports = many tagged VLANs (uplinks).
- Inter-VLAN traffic must route — router-on-a-stick or L3 switch SVIs.
- Harden: explicit port modes, prune trunks, keep the native VLAN unused.

Keep exploring

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